

Economics S5040: Dynamic Stochastic Modeling

Lecture 9 — Job Search, Continuation Values, and Optimal Stopping

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1 Review: What Is a Dynamic Program?

Last week we introduced the main technique of the course: dynamic programming.

A dynamic program has five elements:

1. a **state space**;
2. a **control space**;
3. a **reward function**;
4. a **transition rule**;
5. a **discount factor**.

The value function answers the question

How good is it to be in the current state,
knowing that I will behave optimally in the future?

The policy function answers a different question:

What should I do in each possible state?

The Bellman equation combines the payoff from the current decision with the value of the future:

$$V(x) = \max_u \{r(x, u) + \beta E[V(x') \mid x, u]\}.$$

Today we apply exactly these ideas to a classic economic problem: job search.

2 The McCall Job Search Model

Consider an unemployed worker who receives a wage offer w .

The worker has only two choices:

Accept or Reject.

If the worker accepts, the job lasts forever and pays wage w every period.

If the worker rejects, the worker receives unemployment compensation c this period and draws a new wage offer next period.

Future income is discounted by

$$0 < \beta < 1.$$

Wage offers are independent draws from a distribution F .

The worker wants to maximize expected discounted lifetime income.

3 A New Kind of State Space

This model gives us an important contrast with the finite-state problems from last week.

Suppose wage offers can take any value in an interval

$$W = [w_{\min}, w_{\max}] \subset \mathbb{R}.$$

Then the state is the current wage offer w , and the state space is

$$W = [w_{\min}, w_{\max}].$$

This state space contains a **continuum** of possible states.

It is therefore infinite—indeed, uncountably infinite.

By contrast, the control space contains only two elements:

$$U = \{\text{Accept}, \text{Reject}\}.$$

Thus we have

an infinite, continuous state space
but a two-element control space.

The Bellman principle does not depend on the state space being finite. The same logic we used last week continues to apply.

For computation, we will often replace a continuous wage distribution by a fine finite grid. This is a numerical approximation to the underlying continuous-state problem, not a change in the economic model.

4 The Five Elements of the McCall Problem

We can now identify the five elements:

State:	current wage offer $w \in W$,
Control:	Accept or Reject,
Reward:	w if employed; c while unemployed,
Transition:	after rejection, draw $W' \sim F$,
Discount factor:	β .

Notice that the worker does not choose the wage.

Nature draws the wage. The worker chooses what to do after observing it.

5 The Value of Accepting

Suppose the worker accepts an offer w .

The job is permanent, so lifetime income is

$$w + \beta w + \beta^2 w + \dots .$$

This is a geometric series. Hence the value of accepting is

$$A(w) = \frac{w}{1 - \beta}.$$

This value is increasing in w .

A higher current offer makes stopping the search more attractive.

6 The Value of Rejecting

Now suppose the worker rejects w .

The worker receives c today.

Next period, a new offer W' is drawn, and the worker again behaves optimally. Hence the value of rejecting is

$$C = c + \beta E[V(W')].$$

We will call C the **continuation value**.

This object deserves special attention.

7 Continuation Value

The continuation value is the value of keeping the dynamic problem alive.

In the McCall model,

$$C = c + \beta E[V(W')]$$

means:

1. reject today's offer;
2. receive c today;
3. obtain another wage draw tomorrow;
4. behave optimally from that point onward.

Thus

Continuation value = the value of not stopping yet and preserving the opportunity to choose again tomorrow.
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Continuation values will appear repeatedly in this course.

Dynamic programming works because the value function summarizes all of the future consequences of continuing.

8 Why the Continuation Value Is Especially Simple Here

Wage offers are IID.

If the worker rejects today's wage, tomorrow's offer is a fresh draw from the same distribution F .

Therefore tomorrow's distribution does not depend on today's offer.

As a result,

$$C = c + \beta E[V(W')]$$

does not depend on the current wage w .

This is crucial.

The value of accepting,

$$A(w) = \frac{w}{1 - \beta},$$

risers with w .

The value of continuing,

$$C,$$

is constant with respect to w .

That simple geometry will generate the reservation wage.

9 The Bellman Equation

The worker chooses whichever alternative has the higher value:

$$V(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta E[V(W')] \right\}.$$

Equivalently,

$$V(w) = \max\{A(w), C\}.$$

This equation contains the entire economic problem.

At each wage offer, ask:

$$\text{Is it better to stop, or is it better to continue?}$$

10 The Reservation Wage

Because $A(w)$ is increasing and C is constant, there is a cutoff wage $\bar{w}$ at which the worker is just indifferent:

$$A(\bar{w}) = C.$$

Hence

$$\frac{\bar{w}}{1 - \beta} = c + \beta E[V(W')].$$

Therefore

$$\bar{w} = (1 - \beta) [c + \beta E[V(W')]].$$

The optimal policy is

$$\begin{cases} \text{Reject,} & w < \bar{w}, \\ \text{Accept,} & w \geq \bar{w}. \end{cases}$$

The number $\bar{w}$ is called the **reservation wage**.

Although the state space contains infinitely many possible wage offers, the entire optimal policy can be summarized by one number.

11 Job Search as an Optimal Stopping Problem

The McCall model is an example of an **optimal stopping problem**.

At each date there are two possibilities:

$$\boxed{\text{STOP}} \quad \text{or} \quad \boxed{\text{CONTINUE}}.$$

Stopping means accepting the current wage.

Continuing means rejecting it, collecting unemployment compensation, and waiting for another draw.

The worker's problem is therefore not

“What wage should I choose?”

but

$\boxed{\text{“When should I stop searching?”}}$

The reservation wage tells us exactly when stopping is optimal.

12 A Graph Worth Remembering

Imagine wage w on the horizontal axis and lifetime value on the vertical axis.

The acceptance value

$$A(w) = \frac{w}{1 - \beta}$$

is an upward-sloping straight line.

The continuation value

$$C = c + \beta E[V(W')]$$

is a horizontal line.

They intersect at $\bar{w}$.

To the left of $\bar{w}$,

$$A(w) < C,$$

so the worker rejects.

To the right,

$$A(w) > C,$$

so the worker accepts.

Thus the reservation-wage rule follows directly from the shape of the two value functions.

13 What to Remember

1. Dynamic programming works with continuous as well as finite state spaces.
2. In the McCall model, the wage offer is the state.
3. The state space can be a continuum even though there are only two controls.
4. The continuation value is the value of rejecting today and retaining the opportunity to choose optimally tomorrow.
5. The worker compares the value of stopping with the value of continuing.
6. This comparison produces a reservation wage.
7. The McCall model is an optimal stopping problem.

Next time we will compute the continuation value and reservation wage and ask how they change when the economic environment changes.